WEB 2890 – Final Project Report

WEB 2890 – Brook Brown

1. Overview

For my second WEB 2890 project, I designed and developed a fully responsive IT Support Ticket System prototype for Weber State University. The site includes a modern homepage, contact/help page, login authentication logic, and two dashboards (User and Admin). My goal was to create a system that felt professional, intuitive, and aligned with Weber State's branding guidelines.

2. What I Am Most Proud Of

The aspect I'm most proud of is the overall cohesion and polish of the design. Reviewers consistently mentioned that the layout feels clean, modern, and unified, and that the site closely resembles Weber State's official branding. They also noted that the navigation is intuitive and easy to follow, and that the login routing and dashboard structure function smoothly.

3. What I Could Have Done Better With More Time

With more time, I would have expanded the dashboard ticket views to include full message threads for both admins and users, as suggested by one of the reviewers. I also would have improved the backend logic or integrated a mock API to better simulate real ticket creation, assignment, and messaging. Additionally, I would have enhanced accessibility features, such as adding skip-navigation links and performing more extensive keyboard-only navigation testing.

4. Changes Made in Response to Feedback

In response to the feedback I received, I made several improvements across the project to enhance consistency, usability, and clarity. One reviewer pointed out that the gradients on the contact page didn’t match the rest of the site, so I updated the cards, buttons, and hero section to use a unified gradient style. I also fixed footer spacing and layout issues by adjusting padding and improving responsiveness. Additionally, since one reviewer mentioned needing more information for users who have login issues, I added a more informative description of who can login along with improved error handling, validation messages. These updates helped strengthen the overall usability and cohesiveness of the project.

5. Resources, Assets, and Tools Used

This project relied on several frameworks, libraries, and tools to create a responsive, polished, and maintainable design. I used React as the core structure of the application along with React Router to handle page transitions and dashboard routing. Bootstrap 5 provided the responsive grid system, utility classes, and component foundations, while SCSS allowed me to implement advanced styling through variables, mixins, gradients, and custom theming. I incorporated Bootstrap Icons for clean and consistent iconography, and I created custom SCSS variables to manage colors, shadows, and typography across the site.

6. Justification of Key Design Decisions

Several core design decisions shaped the final look and functionality of the project. I adopted Weber States branding by using purple primary colors and light neutral backgrounds, which reviewers immediately recognized as consistent with WSU design. To maintain a clean and minimal layout, I intentionally avoided visual clutter, kept instructions simple, and used generous whitespace to guide the user's eye and improve readability. A card-based interface was used throughout the home, contact, and dashboard pages to create visual consistency and break content into digestible sections. For the dashboards specifically, I implemented a sidebar navigation layout because it closely mirrors the structure used in real IT admin systems, supports scalable additions like ticket modules or message threads, and adapts well to mobile devices through a collapsible design.

7. Ways I Customized the Framework Design

The project involved extensive SCSS customization to create a cohesive and polished look. I replaced all default Bootstrap colors with custom SCSS variables and built tailored hover states, shadows, and transitions. Form inputs were enhanced with focus rings and smooth animations, while core UI components were fully restyled to match the project theme. On the React side, I developed custom components such as the DashHeader side menu and implemented login validation with conditional routing. Reusable UI elements were created to maintain visual consistency throughout the site. Additionally, responsive adjustments went beyond standard Bootstrap breakpoints, ensuring mobile-friendly layouts for cards, side menus, and grids while maintaining a polished appearance on larger screens.

8. Advanced Features Incorporated

The project implements robust login routing logic, where emails ending with specific domains are directed to the User Dashboard, while emails matching an admin list route to the Admin Dashboard. Input validation handles empty fields, invalid email formats, unrecognized domains, and unsafe characters. The dashboard features a responsive, collapsible side menu ensuring usability across devices. Dynamic UI effects enhance interactivity, including animated card hover-lifts, fade-in shadows, gradient backgrounds, and smooth form focus transitions. The SCSS architecture is designed for reusability, incorporating color variables, mixins, and a nested component structure to maintain consistent styling throughout the application.

9. Problems Experienced During Development

During development, several challenges arose. Adjusting the dashboard side menu spacing required overriding Bootstrap defaults to position navigation closer to the header. Cards with gradient backgrounds initially caused box-shadow clipping, which was resolved through layout adjustments and careful overflow handling. Routing errors occurred when /user-dash and /admin-dash paths were missing from the router configuration. On the homepage, grid alignment issues caused cards to break rows at tablet sizes until the link structure around columns was corrected. Finally, reviewers requested consistent gradient styling across pages, which required SCSS refactoring and updates at the component level to ensure a unified visual design.

10. Conclusion

Overall, this project demonstrates my ability to design, code, and polish a multi-page, responsive, React-based website using professional-level UI/UX principles. I incorporated feedback from multiple reviewers, improved weak points, and delivered a cohesive system that represents both my technical skills and design abilities. This project is portfolio-ready and showcases strong front-end development, thoughtful design, and problem-solving skills.